

IILM UNIVERSITY

School of Computer Science and Engineering



SUMMER INTERNSHIP

7 DARTER INFO SERVICES PRIVATE LIMITED

Software Developer Intern

Name – ANJALI

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B.Tech CSE

15 June – 27 July | New Delhi | Hybrid



Company Profile

7 Darter Info Services Private Limited

Company

An info-tech company focused on real-time technology solutions, consultancy/advisory services and software products.

Focus Areas

Defense • Space & Aviation • Cybersecurity • Integrated & Wireless Communication • Software Development

Positioning

Incorporated under the Government of India's 'Atmanirbhar Bharat' and 'Startup India' initiatives, with an emphasis on Make in India solutions.

Internship context

- Role: Software Developer Intern (SDE)
- Duration: 15 June – 27 July
- Location: New Delhi • Hybrid

Team & Approach

A team of skilled experts and developers applying practical, real-time experience to client problem statements.

Internship Role & Objectives

Software development focused on computer vision and automation

Role

Worked as a Software Developer Intern on a Python-based solution for automated vehicle identification using video/CCTV input.

Primary Objective

Replace manual vehicle tracking with an automated pipeline that detects, validates and reports vehicle number plates.

Objectives

- Build a functional web interface for video upload and processing
- Apply OpenCV-based image processing and EasyOCR for plate recognition
- Validate detected plates against a predefined format
- Produce a usable processed-video output for review and download

Problem Statement & Project Overview

Automating number-plate detection from video footage

Problem

Manual identification and tracking of visitor vehicles from CCTV/video footage, particularly in controlled environments, is time-consuming and difficult to scale.

Proposed Solution

A Flask-based web application that accepts video uploads, detects candidate plates, reads plate text using OCR, validates results and highlights detections in the output video.

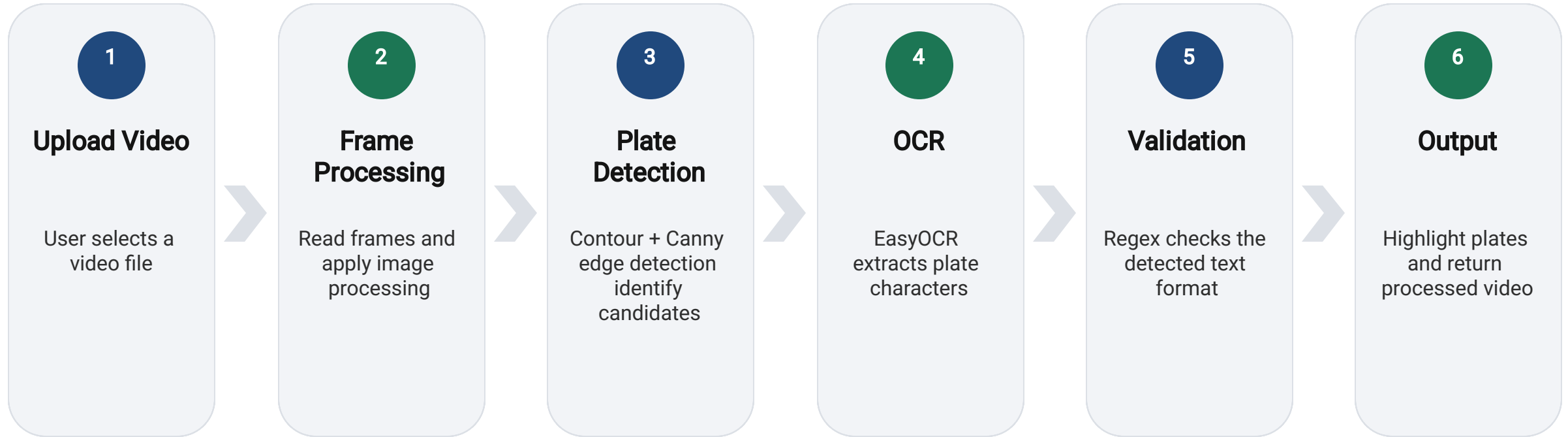
Core Features

- Video upload
- Automatic plate detection
- OCR with EasyOCR
- Regex validation
- Highlighted output video
- Most common valid plate result

Project: Vehicle Number Plate Detection System

Methodology / System Workflow

From uploaded footage to a validated number plate



Validation pattern used: `^[A-Z0-9]{7}$`

Technology Stack & Implementation

Tools used to build and process the application

Python

Application programming

Flask

Web application / backend

OpenCV

Video & image processing

EasyOCR

Optical character recognition

HTML5 + CSS3

User interface

Regex

Plate-format validation

Implementation highlights

Secure upload & file handling • frame-level plate highlighting • processed-video preview/download

Work, Contributions & Results

What was built and what improved

Key Contributions

- Contributed to the Python/Flask application
- Implemented video-processing and number-plate detection logic
- Integrated OpenCV and OCR workflow
- Implemented regex-based validation
- Produced highlighted output and valid-plate reporting

Reported Impact

30%

faster detection

25%

higher accuracy

Reported in the internship/project description supplied for this presentation.

Outcome

A more automated workflow for identifying and validating vehicle plates from footage.

Challenges & Learning Outcomes

Practical engineering lessons from the internship

Challenges

- Video/frame processing complexity
- OCR inaccuracies on difficult frames
- Plate-format validation
- Integrating multiple processing steps into one web workflow
- File handling and output generation

Learning Outcomes

- Python and Flask application development
- OpenCV-based image/video processing
- OCR integration with EasyOCR
- Regex-based data validation
- Debugging and testing a real application
- Translating a manual process into an automated workflow

Professional takeaway: engineering is less about making one algorithm work and more about making the whole pipeline work reliably.

Conclusion & References

Summary of the internship experience

Conclusion

The internship provided practical exposure to software development through a Python/Flask application for automated vehicle number-plate detection. The work combined video processing, computer vision, OCR and validation into a usable web workflow.

Key Takeaway

The project demonstrated how computer-vision components can be integrated into an end-to-end application to reduce manual effort.

References

- 7 Darter Info Services Private Limited – LinkedIn company profile
- Project description and implementation details supplied for the internship presentation

Thank You

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